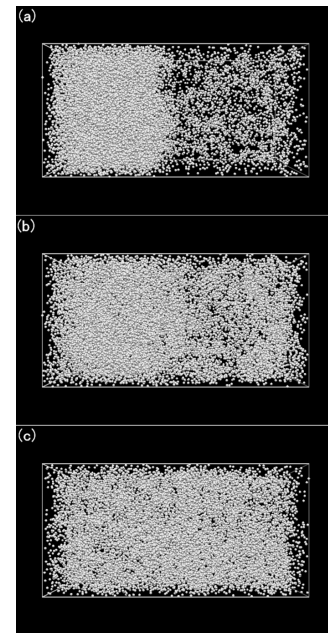


Cell-list algorithm (Phase diagram of the LJ fluid)

- Modify the MD simulation program to include the cell-list algorithm.
- Increase the number of atoms to 1000
- Define an elongated simulation box where the length in the X direction is 2x larger than in the other two directions. Scale the box so that the liquid phase occupies approximately half of the box.
- Characterize the improvement in performance.
- Simulate the system at several different temperatures between 0.4 and 2.
- Determine the local density distribution as a function of the temperature.
- Define and use Binder cumulants to identify the critical temperature.

<https://doi.org/10.1063/1.4720089>



Source: Watanabe et al. J. Chem. Phys. 136, 204102 (2012)